A RING MAP WITH A SECTION HAS AN EXACT EXTENDED COCHAIN COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the extended cochain complex by label, without displaying it; the display defining it, together with the prose fixing the degrees, is delivered as a context span so that the reader can see the object, and it is not counted as a statement.

Editorial note. The proof displays the complex of the constant cosimplicial module as a diagram with alternating identity and zero maps; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The extended cochain complex of a ring map and its degree convention. Same chapter, the display equation-extended-complex together with the surrounding prose, delivered as a context span. Fixes the complex whose exactness is asserted and the degrees in which its terms sit.
- A ring map with a section gives a homotopy equivalence from the constant cosimplicial algebra. Simplicial, Lemma simplicial-lemma-push-outs-simplicial-object-w-section, invoked at proof lines 748-750. Produces the homotopy equivalence that the rest of the proof transports.
- Functors preserve homotopies of cosimplicial objects, and a homotopy equivalence of cosimplicial objects induces a homotopy equivalence of associated complexes. Simplicial, Lemma simplicial-lemma-functorial-homotopy at proof lines 757-758 and Simplicial, Lemma simplicial-lemma-homotopy-s-Q at proof lines 760-761. Carries the homotopy equivalence through tensoring with M and then through the associated complex functor.

Source reference. Original equation 023K: sources/stacks-descent/pinned-original-source.tex, source lines 729–729. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 019Y: sources/stacks-chow/tags.txt, source lines 1669–1669. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01A0: sources/stacks-chow/tags.txt, source lines 1671–1671. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 019Z: sources/stacks-chow/tags.txt, source lines 1670–1670. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Suppose that $R \to A$ has a section. Then for any R-module M the extended cochain complex (023K) is exact.*

Proof. By Simplicial, Lemma 019Z the map $R \to (A/R)_\bullet$ is a homotopy equivalence of cosimplicial R-algebras (here R denotes the constant cosimplicial R-algebra). Hence $M \to (A/R)_\bullet \otimes_R M$ is a homotopy equivalence in the category of cosimplicial R-modules, because $\otimes_R M$ is a functor from the category of R-algebras to the category of R-modules, see Simplicial, Lemma 019Y. This implies that the induced map of associated complexes is a homotopy equivalence, see Simplicial, Lemma 01A0. Since the complex associated to the constant cosimplicial R-module M is the complex

$$M \xrightarrow{0} M \xrightarrow{1} M \xrightarrow{0} M \xrightarrow{1} M \ldots$$

we win (since the extended version simply puts an extra M at the beginning). $\square$